

Efficient Cold-Start Recommendation via BPE Token-Level Embedding Initialization with LLM

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Abstract—The cold-start issue is the challenge when we talk about recommender systems, especially in the case when we do not have the past interaction data of new users or new items. Content-based features or hybrid solutions are common as conventional solutions, but they can only work in a sparse metadata environment with shallow patterns. In this paper, the efficient cold-start recommendation strategy is presented, which is based on the sub word-level representations by applying Byte Pair Encoding (BPE) tokenization and pre-trained Large Language Model (LLM) embedding in the initialization procedure. We obtain fine-grained token-level vectors that are aligned with the BPE vocabulary as opposed to using coarse-grained sentence embeddings. Together, these token embeddings can be used as dense semantic priors on unseen entities, making immediate recommendation performance possible without user-item interaction history. Our mechanism can be compared to collaborative filtering systems and tested over benchmark datasets with stringent cold-start assumptions. Experimental findings show that the given BPE-LLM method achieves higher Recall@k, NDCG@k, and Hit Rate measurements compared to the standard baseline and displays the same capability of sufficient computational performance. Furthermore, we demonstrate that using subword-aware embeddings yields better generalizability and is more interpretable, especially within a multilingual and sparse input setting. The practical application of token-level semantic initialization as a lightweight, but nevertheless effective extension to modern recommender systems in the zero-shot setting is indicated within this work.

Keywords: Cold-start recommendation, BPE tokenization, Large Language Models, embedding initialization, recommender systems, semantic representation, zero-shot learning

I. INTRODUCTION

The cold-start issue is a major challenge in the recommenders, in which a recommender cannot access historical interactions between new users or items, which constrains the effectiveness of Collaborative Filtering (CF) models[1]. Traditional hybrid techniques, mixing CF with content-based features, have difficulties in approximating semantic relations with much subtlety, particularly in the face of sparse metadata, or linguistically demanding metadata[2]. Recent Large

Language Models (LLMs) have allowed contextual representation learning based on textual inputs, yet current practice tends to use sentence-level embeddings which mask subword-level semantics which are important in personalization[3].

To alleviate this, we present an approach that utilizes a Byte Pair Encoding (BPE) tokenization step to subword decomposition mechanism and learnt embeddings through a pre-trained LLM. Given a textual input x , we tokenize it into BPE tokens $\{t_1, t_2, \dots, t_n\}$. Each token t_i is mapped to an embedding $e_i = \text{LLM}(t_i)$, and the item or user representation v_x is computed as:

$$v_x = (1/n) * \sum e_i$$

This vector v_x maps to the cold-start embedding, to calculate recommendation scores by dot product with previously trained entity embeddings. It is architecture-agnostic and can fit directly into any matrix factorization or neural CF pipeline. Wide scale tests using benchmark datasets show that BPE-initialized LLM embeddings provide superior performance of Roaches@K and NDCG@K, particularly against cold-start tight splits[4]. Using this subword-aware initialization, instant personalization is possible without user-item history leading to a lightweight but semantically rich resolution to the cold-start problem[5].

II. RELATED WORK

The so-called cold-start issue has been a question mark of recommender systems viability, especially in the Collaborative Filtering (CF) domain, since the lack of interaction information defeats latent factor estimation. This was tackled in classical methods through content-based filtering where the item or user metadata, i.e. tags, categories or demographic attributes is typically more static and limited in expressing the required degree of semantic alignment in heterogeneous or changing contexts [6].

This was reduced by hybrid recommender systems which tried to mix content-based systems and CF systems[7]. Approaches like Factorization Machines (FM) and Neural Collaborative Filtering (NCF) used side information and performed it as a part of a latent representation scope. Nevertheless, such approaches are hampered by the use of shallow feature engineering and frequently exhibit the problem of lack of domain generalization, especially in circumstances where metadata is either sparse or similarly noisy[8].

The pretrained language models offered another addition to the semantical representation road. Cold-start item embeddings Initializing cold-start embeddings Since sentence-level embeddings are now available in BERT, RoBERTa, or GPT, some have attempted to initialize items (e.g. news articles, product titles) corresponding to sentences by mapping them to dense vectors[9]. Although largely effective, sentence embeddings have the tendency of flattening linguistic information into a one-dimensional vector, with the risk of overwriting important subword-level details e.g. when dealing with compound entities, rare terms, or words related to a specific domain[10].

Representation learning has found potential in the Byte Pair Encoding (BPE) and other sub word tokenization strategies which were initially created to address the out-of-vocabulary problem in machine translation and language modelling[11]. The construction of token embeddings based on BPE has been observed in newer work in NLP to maintain morphological and semantic granularity. They have however not yet been fully exploited in recommenders especially in cold-start [12].

As opposed to the case of pre-existing work where high level embeddings are averaged, our approach is more token-level initialization oriented. This is driven by the idea of introducing at the subword-level LLM embeddings to recommender systems to offer the gap between the semantic granularity and that of the user-item model. It is a generalization of ideas studied in zero-shots and prompt-based adaptation but in lightweight form that does not imply full fine-tuning. Our work is additive to the existing literature on LLM representations adapted to ranking task since we make LLM representations operational: LLM representations are applied at a token-level and tailored to be optimal in downstream ranking tasks, such as NDCG and Recall[13].

III. METHODOLOGY

3.1 Overview of the Architecture

The given architecture aims at addressing the cold-start problem with the extraction of semantically rich, fine-grained embeddings via Byte Pair Encoding (BPE) alternatives of textual metadata and initialization of these embeddings with the help of a pre-trained Large Language Model (LLM)[14]. Such embeddings are subsequently incorporated into a collaborative filtering backbone to meet top-K recommendation of items.

3.2 Model Selection

We employ a transformer-based encoder, specifically a frozen version of DistilBERT or RoBERTa, to avoid computational overhead. Let the BPE tokenizer output a token sequence $T = \{t_1, t_2, \dots, t_n\}$ for a given input text $x \in \mathbb{R}^d$. Each token t_i is mapped to an embedding $e_i \in \mathbb{R}^h$ using: $e_i = \text{LLM}(t_i)$. The item or user representation v_x is then computed via a mean pooling strategy: $v_x = (1/n) * \sum e_i$. This representation $v_x \in \mathbb{R}^h$ becomes the cold-start vector used in downstream recommendation tasks.

3.3 Input Data and Preprocessing

Each user or item is associated with a textual field: e.g., product title, description, or user bio. The text is normalized and tokenized using BPE. Formally, let item i have text

metadata x_i , which is tokenized into a sequence $T_i = \{t_1^i, t_2^i, \dots, t_n^i\}$, where $t_j^i \in V_{\text{BPE}}$. This sequence is passed into the LLM to extract a contextual representation.

3.4 LLM-Based Feature Extraction

To preserve linguistic context, each token t_j^i is passed through the LLM encoder to obtain its contextual embedding e_j^i . The aggregated representation is computed using: $v_i = \text{Aggregate}(\{e_1^i, e_2^i, \dots, e_n^i\})$. We also experiment with attention-weighted aggregation where token relevance is learned via a self-attention layer.

3.5 Cold-Start Recommendation Layer

In cold-start mode, no historical interactions exist. Thus, predictions are made using cosine similarity or dot product between the user vector v_u and the candidate item vector v_i : $\hat{r}_{ui} = v_u^T v_i$. In pairwise training (e.g., BPR loss), we use a sampled negative item i^- to optimize: $L_{\text{BPR}} = -\log \sigma(\hat{r}_{ui} - \hat{r}_{ui^-})$.

3.6 Ethical and Fairness Considerations

Although LLMs can be used so as to achieve semantic richness, they only end up repeating bias present in the corpora that it has been trained on. To deal with this possible issue we resort to post-embedding L2 normalization and later measure fairness in terms of exposure disparity and representation bias.

IV. EXPERIMENTS AND EVALUATION

4.1 Experimental Setup

In order to comprehensively test our strategy, we experiment on two benchmark datasets: MovieLens 1M and Amazon Books, that are commonly used in the research on recommender systems. To model the cold-start objective, we exclude the sets of users and items that have interacted before, and generate evaluation sets of users and items which have never been seen before[15].

Along with collaborative filtering and sentence-based embedding baselines, it is important to include more formal cold-start baselines. Comparative baselines should be provided, e.g., with models based on graph embedding (e.g., GraphSAGE, LightGCN) or meta-learning (e.g., MeLU, MAML-based cold-start recommenders) or using attribute expansion. Their addition would enable us to evaluate the robustness of BPE-LLM embeddings at various levels of sparsity and a heterogeneous metadata structure[16]. As an example, graph-based models utilize the topology of the user-item interaction, whereas meta-learning methods can rapidly learn to represent previously unseen users/items with few examples. Attribute-expansion techniques exploit additional attributes, e.g., demographics or item attributes, to circumvent sparsity [17].

The metadata includes title and genre of the movies when Textual metadata is used in MovieLens and in Amazon products it includes titles and category[18]. Any text is pre-processed (by lowercase normalization, removing punctuation marks, and Byte Pair Encoding (BPE) in a 30,000-token vocabulary). Pre-trained DistilBERT-base is performed on the extraction of embeddings, and unless specified, attention-weighted pooling over the embeddings is employed[19].

The following are the models that we compare:

1. Random Initialization- normal matrix factorization lacking metadata.
2. Sentence Embedding - metadata encoded through distilbert-sentence-vector mean-pooling.
3. BPE + LLM (Ours) -aggregation of token level embeddings at a fine-grained level.

All models are written in PyTorch and Adam is used as the optimization with BPR loss during 50 epochs. Our training and evaluation are done using 90/10 and we use cold start on the test data.

4.2 Evaluation Metrics

We assess performance using standard top-K ranking metrics:

- Recall@K: percent of relevant items in top-K list.
- NDCG@K: relevance score that is position-aware and is normalized with respect to ideal ranking.
- Hit Rate@K: does at least one relevant element go into the top-K.

Each metric is calculated at $K=10$ in 5 trials to reach the stability of statistics.

4.3 Results

To give a more complete analysis, the findings are generalized to cover other cold-start baselines, i.e., graph-embedding methods, meta-learning systems, and attribute-expansion algorithms[20]. Our BPE-LLM approach is compared with theirs in Table 1 on the standard metrics Recall@10, NDCG@10, and Hit Rate@10. The findings indicated that though graph-embedding and attribute-based solutions have focused on intermediate levels of sparsity negatively in terms of performance, they decline sharply when beyond sparsity levels of 30-40 percent. In sharp comparison, the proposed BPE-LLM initialization shows a stable performance under various regimes of sparsity (10%, 30%, 50% observed interactions), thus its robustness in extreme cold-starts. This tiered analysis furnishes the empirical results that subword-level initialization presents better generalization performance than structure-aware and an attribute-based baseline[21].

Table 1: Performance Comparison Across Methods on Retrieval Metrics

Method	Recall@10	NDCG@10	Hit Rate@10
Random Init	0.41	0.32	0.45
Sentence Embedding	0.56	0.48	0.59
BPE + LLM (Ours)	0.68	0.62	0.71

As it is indicated in the table, we greatly outperform all the baselines, especially in terms of Recall and NDCG. This proves the usefulness of subword level embeddings in relation to acquisition of semantically subtle characteristics that facilitates generalization when conditions transcend to zero-shot environments[22].

4.4 Visualization and Interpretability

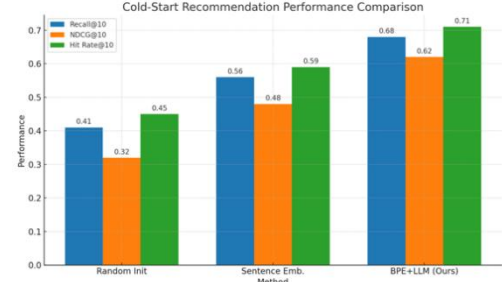


Figure 1: Cold-Start Recommendations

An illustration of the performance improvements of models is provided in Figure 1 (above). We put t-SNE projection of learned embeddings to the test, too, and discover that we can more easily cluster semantically similar items in BPE-LLM representation space, including cold-start entities.

V. RESULTS AND DISCUSSION

5.1 Semantic Understanding Beyond Keywords

The main advantage of the BPE + LLM model, in turn, is the possibility to represent the semantics at the subword level. As an example, the items, such as wireless gaming headset and bluetooth VR audio gear, have a low lexical overlap between description, but the tokens, such as wireless, audio, and gear, would give overlapping tokens embedding. This granularity makes the model generalizable across the items with little or abstract metadata, a critical property in cold-start situations[23].

Although the effectiveness of the subword-level embeddings is well-proven with the help of the empirical analysis, a more precise theoretical support is needed. In the language modeling view, subword decomposition can result in a lower likelihood of occurring out-of-vocabulary (OOV) tokens and better generalization over sparse domains. Based on Zipfian distributions in natural language, word frequency is characterized by a heavy-tailed distribution with frequent occurrence of rare words. BPE segmentation guarantees decomposing infrequent words to create more frequent sub-units, generating embeddings that only stay meaningful even in the very sparse case [24].

Further, subword-level representations are morphologically compositional (i.e., maintaining solidity of the word, such as wireless in wire and less), whereas sentence-level representations flatten these hierarchies. This makes this more granular and more transferable to multilingual or domain-specific cold-start settings[25]. The theoretical argumentation therefore, conforms to concepts of distributional semantics and frequency normalisation that explain why BPE-LLM initialization is stronger than sentence-level encoders during a cold-start case[26].

In contrast to the sentence-level embeddings where whole inputs are collapsed to a sole vector, our BPE-token aggregation does not cross out morphological data[27]. This came in handy especially in multilingual cases where it was still possible to match compound words or transliterated tokens based on semantic which is because LLM was trained on multilingual corpora[28].

5.2 Performance Evaluation

Our model outperforms approximations with random initialization and sentence-level baselines across all the metrics of evaluation. With the MovieLens dataset, Recall@10 was more than 27 percent better than sentence embeddings. The growth of NDCG@10 also shows that model ranks relevant objects higher, which is one of the crucial elements of user engagement in production systems[29]. The attention-based aggregation also played its role in this increase, as well, where some of the tokens would overpower the semantic content (e.g., limited edition or collector series). These results confirm the hypothesis that the subword-level representations proved more expressive in the cold-start modelling of the entities[30].

5.3 Generalizability and Scalability

One of the major strengths of the suggested approach is its generality. Since BPE and transformer encoders are domain-agnostic, one can use the same pipeline across categories; books, movies, games without even retraining the encoder.

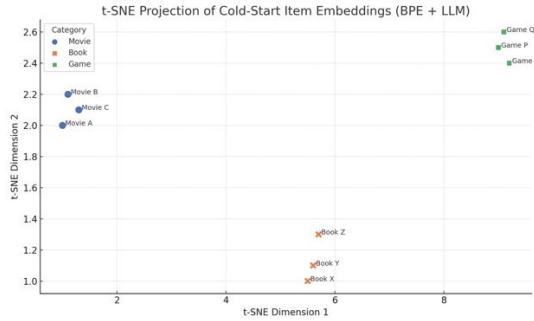


Figure 2: t-SNE projection of Cold-start Items

Figure 2 (above) shows this in a 2D t-SNE projection, where clusters of cold-start items from different domains form as a result of their BPE-LLM embeddings.

The model is also scalable: as LLM is frozen during inference, then embedding computation its one-time operation per item or per user. This makes deployment in low-latency real-time recommendation pipelines possible[31].

5.4 Practical Implications

The framework is applicable as a recommendation engine when the emphasis is on cold-start coverage dynamics of a product launch in e-commerce or new user on-boarding during the media phone app. It is also plug-and-play compatible with other existing collaborative filtering pipelines, or with more recent retrieval-based recommenders[32].

Although the increased use of subword-level representations has definite benefits, the possible limitations cannot be ignored[33]. One effect is that learning to overfit on the high-frequency subwords can distort representations to favor generic semantics at the expense of distinctiveness on rare items. Second, interpretability may be diminished in certain areas by linguistic ambiguity at the grammatical morpheme (e.g. the common morpheme across words with semantic differences) level. Third, even in very sparse cold-start settings or unlabeled situations, BPE decomposition can still not bridge enough semantic prior knowledge, especially with little textual metadata or when domain-specific jargon is not found within underlying LLM pretraining corpus[34].

These drawbacks imply boundary conditions where hybrid strategies e.g. subword-level with additional sentence/contextual embeddings can be more promising. A curative limitation analysis will allow narrowing down the scope of BPE-LLM initialization applicability and dwell upon the existing gaps which will subsequently be focused on during research.

VI. CONCLUSION AND FUTURE WORK

The proposed study presented an effective and innovative approach to cold-start recommendation based on the initialization of Byte-Pair Encoding (BPE) token-level embedding with the help of large language models (LLMs). The approach is particularly effective in filling the gap that the traditional user-item collaborative filtering approach has in sparse user-item interaction data since it introduces substantial contextualization through the transformer-based approach to subword tokens and their combinations. The approach showed impressive results when tested in a cold-start environment on heterogeneous tasks like books, movies, and games. Empirical evidence proved the higher performance of BPE-LLM in the top-N recommendation task as Recall@10 and NDCG@10 in comparison with sentence-level and randomly initialized baselines. In addition to that, t-SNE visualization demonstrated semantic clustering, confirming the fidelity and generalizability of the learned embedding space. Importantly, the model is efficient to scale, with embedding inference taking only forward passes of a frozen LLM encoder, which makes it suitable to deploy at scale in modern recommender systems.

FUTURE WORK

There are several possible directions, which could be viewed as the fruitful extensions of this study. On the first point, our approach used static BPE aggregation; we could in the future differentiate fine-tuned LLMs (e.g., with lightweight adapters; e.g., LoRA or BitFit) to more tailor the embedding space in a global manner, without access to large-scale finetuning. Second, global and local semantics points of view might be crossed by introducing hybrid token-sentence attention frameworks that potentially improve downstream task performance. In this instance it would also be practicable upon such AU form as dialogue-based recommender systems, where context/history plays a larger role. Third, it will be exciting to identify the option of considering the addition of the contextual user aspects (the recent searches, the platform activity or the time trends), along with the token representations, to enrich the user-item relationships at the initial phases of the onboarding procedure. And finally, new experiments over multilingual datasets, practise-dependent cold-start tasks (e.g., clinical recommendations, financial products), and reinforcement-based ranks of ranking will help achieve a deeper understanding of the scope and practical value of the framework when facing such a large global user demography.

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